

Tree Diagrams and Combined Events

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

A fair coin is flipped twice.

How many possible outcomes are there?

.....

2 [2 marks]

A fair coin is flipped and a fair four-sided dice is rolled.

(a) How many possible outcomes are there?

(b) Write down the probability of getting a head and a 4.

.....

3 [2 marks]

A and B are independent events with $P(A) = 0.5$ and $P(B) = 0.2$

Work out $P(A \text{ and } B)$.

.....

4 [2 marks]

A fair coin is flipped twice.

Work out the probability of getting two heads.

.....

5 [3 marks]

A bag contains 3 red and 7 blue counters. A counter is taken, its colour recorded, and it is **replaced**. A second counter is then taken.

(a) Work out the probability that both counters are red.

(b) Work out the probability that both counters are blue.

.....

6

[3 marks]

Using the same bag and method (3 red, 7 blue, with replacement, two draws), work out the probability of getting at least one red counter.

7

[3 marks]

A spinner has probability 0.3 of landing on 'win'. It is spun twice.
Work out the probability of winning exactly once.

8

[3 marks]

Two fair six-sided dice are rolled.
(a) Work out the probability that both scores are even.
(b) Work out the probability of getting at least one 6.

9

[4 marks]

A bag contains 4 red and 6 blue counters. Two counters are taken at random **without** replacement.
(a) Write down the probability that the first counter is red.
(b) Given that the first counter was red, write down the probability that the second is red.
(c) Work out the probability that both counters are red.

10

[4 marks]

Using the same bag (4 red, 6 blue, two counters taken without replacement), work out the probability that the two counters are of different colours.

1 [2 marks]

A and B are independent with $P(A) = 0.6$ and $P(B) = 0.25$
Work out $P(A \text{ and not } B)$.

2 [2 marks]

A biased coin has $P(\text{head}) = 0.7$. It is flipped three times.
Work out the probability of getting three heads.

3 [3 marks]

The same biased coin ($P(\text{head}) = 0.7$) is flipped three times.
Work out the probability of getting at least one head.

4 [3 marks]

A bag contains 5 red and 4 green counters. Two are taken without replacement.
Work out the probability that they are the same colour.

5 [4 marks]

A factory uses two machines. Machine A makes 70% of the items and 3% of them are faulty. Machine B makes the other 30% and 8% of those are faulty.
Work out the probability that a randomly chosen item is faulty.

6

[4 marks]

Using the same factory information, an item is found to be faulty.
Work out the probability that it was made by machine B.

7

[4 marks]

A bag contains 6 white and 4 black counters. Three are taken at random without replacement.
Work out the probability that all three are white.

8

[4 marks]

A bag contains n blue counters and 4 red counters.

Two counters are taken at random without replacement. The probability that both are blue is $\frac{1}{3}$

(a) Show that $n^2 - 5n - 6 = 0$

(b) Work out the number of blue counters.

9

[4 marks]

A spinner has probability 0.4 of landing on red. It is spun three times.
Work out the probability of getting exactly two reds.

10

[5 marks]

Bag A contains 3 red and 5 blue counters. Bag B contains 4 red and 2 blue counters.
One counter is taken at random from bag A and placed into bag B.
A counter is then taken at random from bag B.
Work out the probability that the counter taken from bag B is red.
